SCH (PCBA) Part Number: DT10965

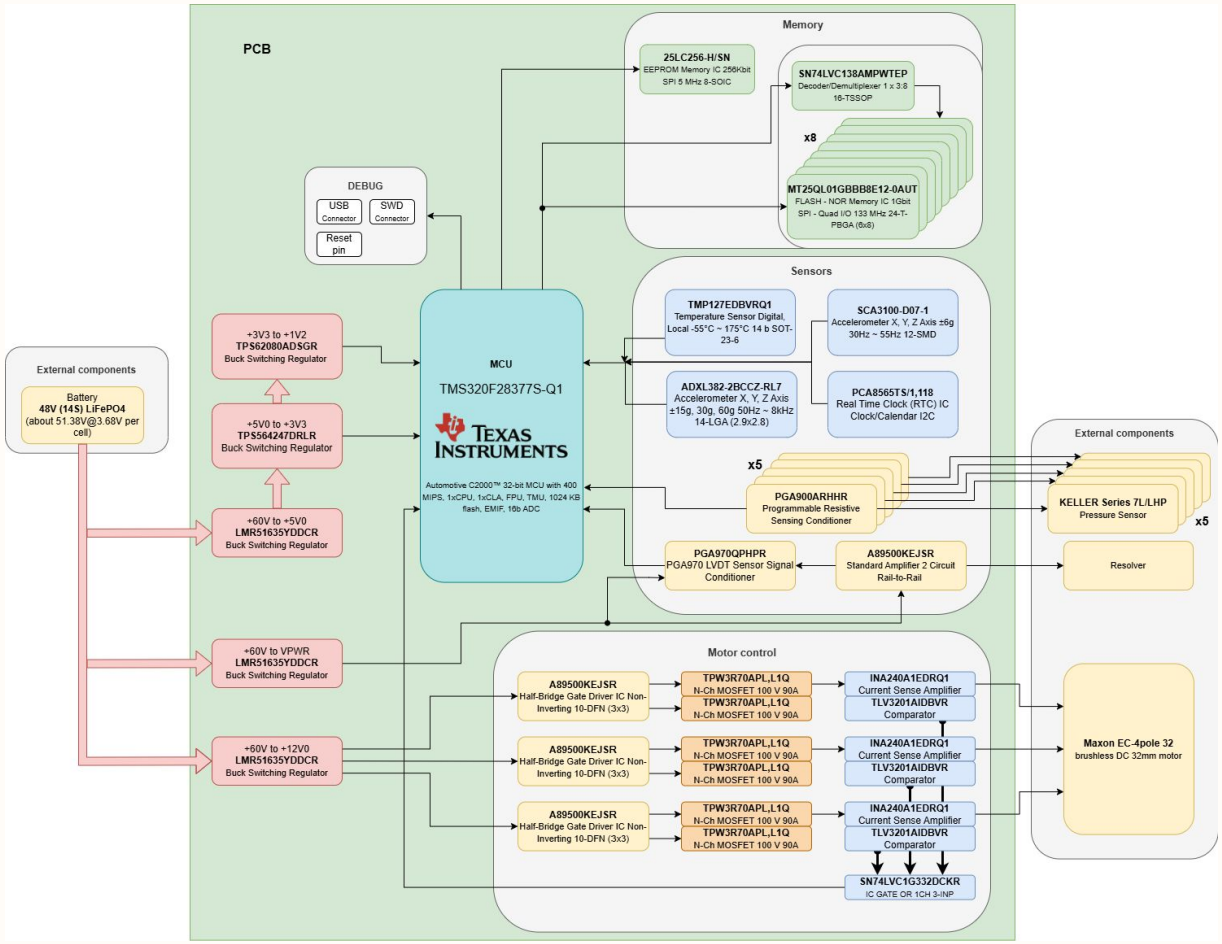
SCHEMATIC MAJOR REVISION HISTORY

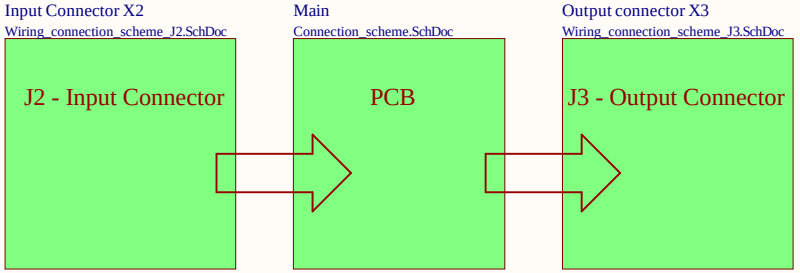
REVISION	SUBREVISION	CHANGE DESCRIPTION	YYYY/MM/DD
V1	00	Initial version	7.07.2025

PCB Part Number: DT10966

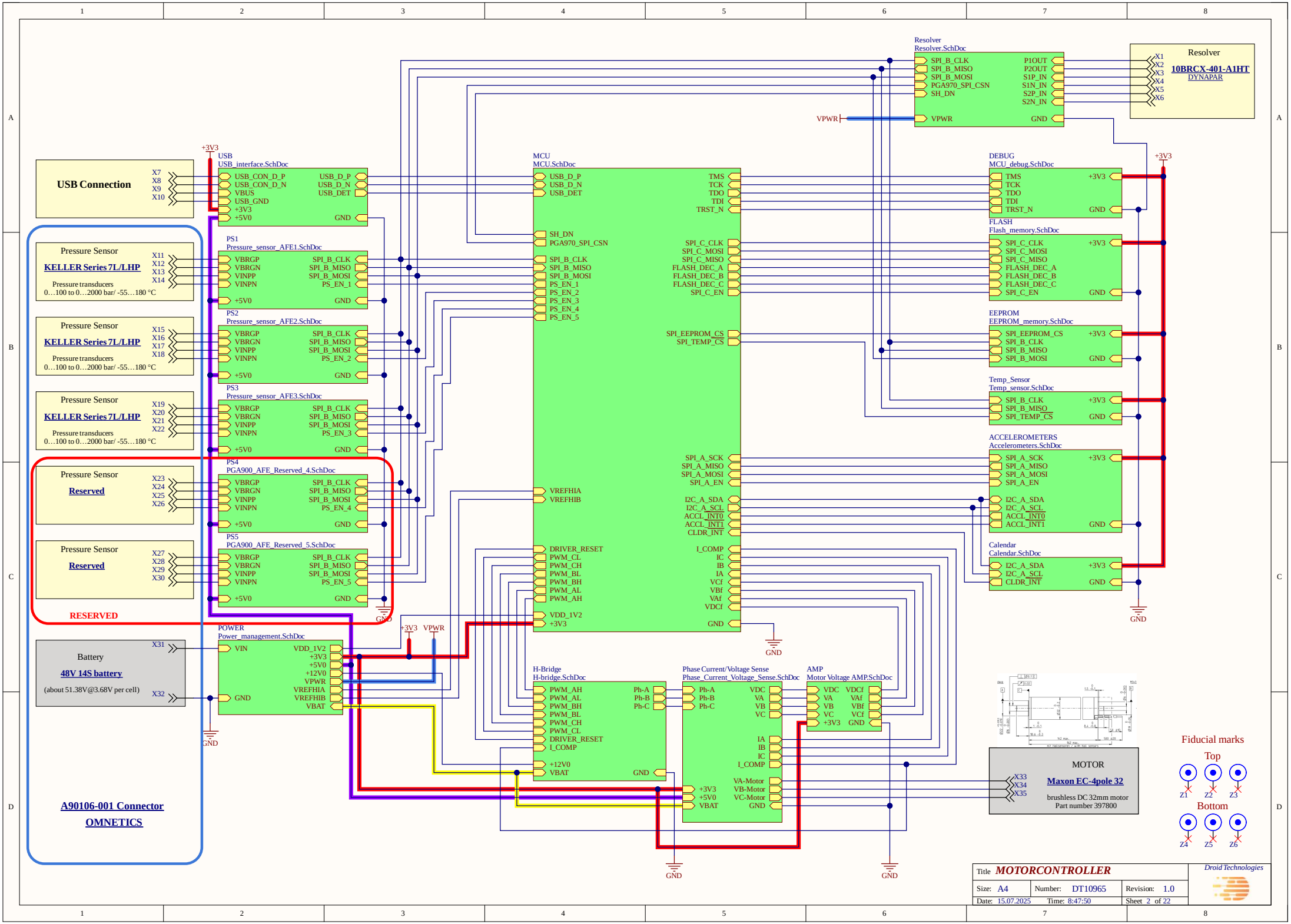
PCB MAJOR REVISION HISTORY

REVISION	SUBREVISION	CHANGE DESCRIPTION	YYYY/MM/DD
V1	00	Initial version	dd.mm.2025

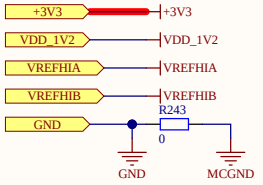


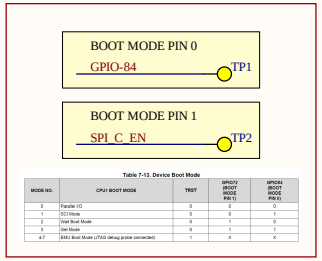


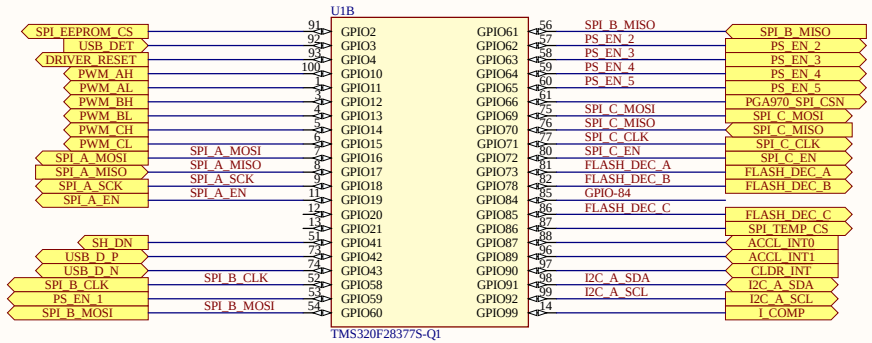
Title	MOTORCONTROLLER		Droid Technologies
Size: A4	Number: DT10965	Revision: 1.0	
Date: 15.07.2025	Time: 8:47:50	Sheet 1 of 22	


USB Connection
X7
X8
X9
X10
USB
USB_interface.SchDoc
USB_CON_D_P
USB_CON_D_N
VBUS
USB_GND
+3V3
+5V0
USB_D_P
USB_D_N
USB_DET
GND
Pressure Sensor
KELLER Series 7L/LHP
Pressure transducers
0…100 to 0…2000 bar/ -55…180 °C
X11
X12
X13
X14
PS1
Pressure_sensor_AFE1.SchDoc
VBRGP
VBRGN
VINPP
VINPN
SPI_B_CLK
SPI_B_MISO
SPI_B_MOSI
PS_EN_1
X15
X16
X17
X18
PS2
Pressure_sensor_AFE2.SchDoc
PS_EN_2
X19
X20
X21
X22
PS3
Pressure_sensor_AFE3.SchDoc
PS_EN_3
Pressure Sensor
Reserved
X23
X24
X25
X26
PS4
PGA900_AFE_Reserved_4.SchDoc
PS_EN_4
Pressure Sensor
Reserved
X27
X28
X29
X30
PS5
PGA900_AFE_Reserved_5.SchDoc
PS_EN_5
RESERVED
Battery
48V 14S battery
(about 51.38V@3.68V per cell)
X31
X32
POWER
Power_management.SchDoc
VIN
VDD_1V2
+3V3
+5V0
+12V0
VPWR
VREFHIA
VREFHIB
VBAT
A90106-001 Connector
OMNETICS
MCU
MCU.SchDoc
USB_D_P
USB_D_N
USB_DET
SH_DN
PGA970_SPI_CSN
SPI_B_CLK
SPI_B_MISO
SPI_B_MOSI
PS_EN_1
PS_EN_2
PS_EN_3
PS_EN_4
PS_EN_5
VREFHIA
VREFHIB
DRIVER_RESET
PWM_CL
PWM_CH
PWM_BL
PWM_BH
PWM_AL
PWM_AH
VDD_1V2
+3V3
GND
TMS
TCK
TDO
TDI
TRST_N
SPI_C_CLK
SPI_C_MOSI
SPI_C_MISO
FLASH_DEC_A
FLASH_DEC_B
FLASH_DEC_C
SPI_C_EN
SPI_EEPROM_CS
SPI_TEMP_CS
SPI_A_SCK
SPI_A_MISO
SPI_A_MOSI
SPI_A_EN
I2C_A_SDA
I2C_A_SCL
ACCL_INT0
ACCL_INT1
CLDR_INT
I_COMP
IC
IB
IA
VCf
VBf
VAf
VDCf
Resolver
Resolver.SchDoc
SPI_B_CLK
SPI_B_MISO
SPI_B_MOSI
PGA970_SPI_CSN
SH_DN
VPWR
P1OUT
P2OUT
S1P_IN
S1N_IN
S2P_IN
S2N_IN
GND
Resolver
10BRCX-401-A1HT
DYNAPAR
X1
X2
X3
X4
X5
X6
DEBUG
MCU_debug.SchDoc
TMS
TCK
TDO
TDI
TRST_N
+3V3
GND
FLASH
Flash_memory.SchDoc
SPI_C_CLK
SPI_C_MOSI
SPI_C_MISO
FLASH_DEC_A
FLASH_DEC_B
FLASH_DEC_C
SPI_C_EN
EEPROM
EEPROM_memory.SchDoc
SPI_EEPROM_CS
SPI_B_CLK
SPI_B_MISO
SPI_B_MOSI
Temp_Sensor
Temp_sensor.SchDoc
SPI_TEMP_CS
ACCELEROMETERS
Accelerometers.SchDoc
SPI_A_SCK
SPI_A_MISO
SPI_A_MOSI
SPI_A_EN
I2C_A_SDA
I2C_A_SCL
ACCL_INT0
ACCL_INT1
Calendar
Calendar.SchDoc
CLDR_INT
H-Bridge
H-bridge.SchDoc
PWM_AH
PWM_AL
PWM_BH
PWM_BL
PWM_CH
PWM_CL
DRIVER_RESET
I_COMP
+12V0
VBAT
Ph-A
Ph-B
Ph-C
GND
Phase Current/Voltage Sense
Phase_Current_Voltage_Sense.SchDoc
VDC
VA
VB
VC
IA
IB
IC
I_COMP
VA-Motor
VB-Motor
VC-Motor
AMP
Motor Voltage AMP.SchDoc
VDC
VA
VB
VC
+3V3
GND
VDCf
VAf
VBf
VCf
MOTOR
Maxon EC-4pole 32
brushless DC 32mm motor
Part number 397800
X33
X34
X35
Fiducial marks
Top
Z1
Z2
Z3
Bottom
Z4
Z5
Z6
Title MOTORCONTROLLER
Size: A4
Number: DT10965
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Sheet 2 of 22
Droid Technologies

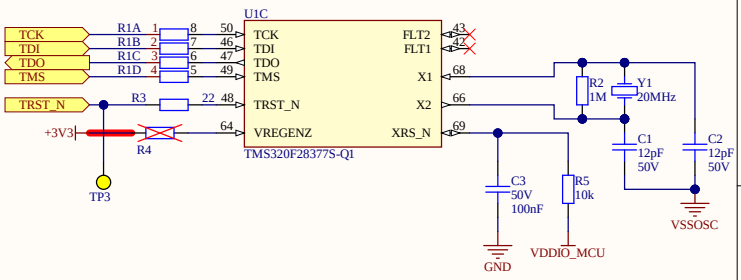
MCU





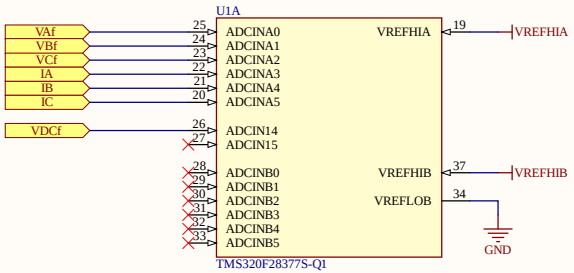


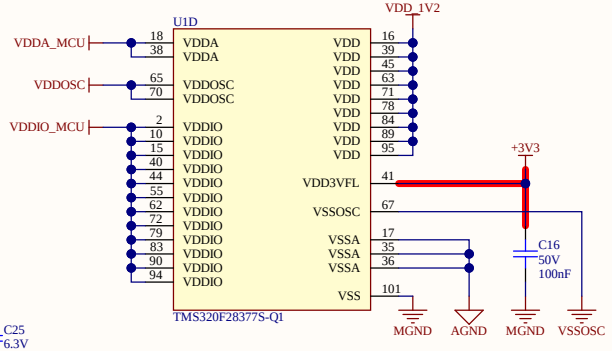
Device supply voltage, I/O, VDDIO = +3.14V to +3.47V (30mA(max))
Device supply voltage, VDD = +1.14V to +1.26V (400mA(max))
Analog supply voltage, VDDA = +3.14V to +3.47V (20mA(max))
IDD3VFL = 40mA(max) @
Ambient operating temperature (max) = –40°C to 125°C free-air

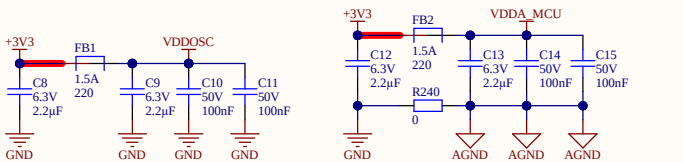


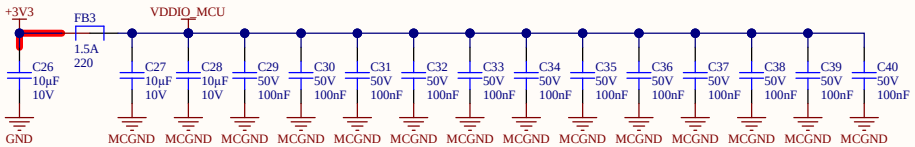
SPI Connection table

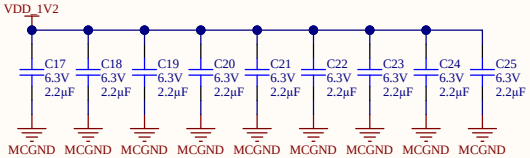
Type	IC connection	Maximum SPI Working Frequency	SPI Port#	SPI CS PIN	
MCU	TMS320F28377S-Q1	3 high-speed SPI-UP to 50 MHz	SPI A/B/C	Port Name	I/O
Flash	MT25QL01GBBB8E12-0AUT	90/27 (fC/fR)- Clock frequency	SPI C	SPI_C_EN	GPIO72(+73,78,86)
EEPROM	25LC256-H/SN	3 MHz (at +3,3V)	SPI B	SPI_EEPROM_CS	GPIO2
TempSens	TMP127EDBVRQ1	10 MHz (Max)	SPI A	SPI_TEMP_\C\S\	GPIO86
Accelerometer	SCA3100-D07-1	8 MHz (MAX)	SPI A	SPI_A_EN	GPIO19
Resolver AFE	PGA970QPHPR	1 MHz (MAX)	SPI B	PGA970_SPI_CSN	GPIO66
Pressure sensor AFE1	PGA900ARHHR	1 MHz (MAX)	SPI B	PS_EN_1	GPIO59
Pressure sensor AFE2	PGA900ARHHR	1 MHz (MAX)	SPI B	PS_EN_2	GPIO62
Pressure sensor AFE3	PGA900ARHHR	1 MHz (MAX)	SPI B	PS_EN_3	GPIO63
Pressure sensor AFE4	PGA900ARHHR	1 MHz (MAX)	SPI B	PS_EN_4	GPIO64
Pressure sensor AFE5	PGA900ARHHR	1 MHz (MAX)	SPI B	PS_EN_5	GPIO65



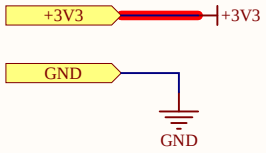


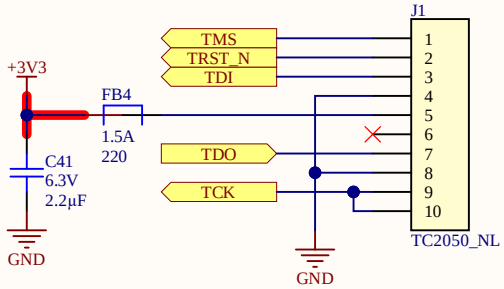






Title	MOTORCONTROLLER		Droid Technologies
Size: A4	Number: DT10965	Revision: 1.0	
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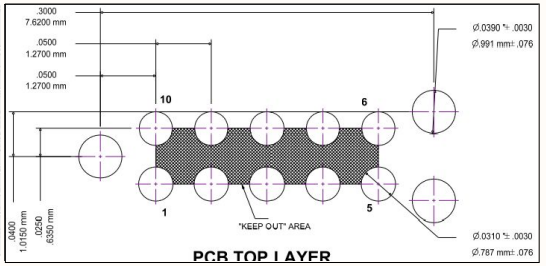




Use TC2050-IDC-430-NL cable for debug connection

TC2050-IDC-430-NL connection scheme

14-pin Ribbon Pin #	TC2050-IDC-430 Pin #
1	1
2	2
...	...
8	8
9	9
10	(NC)
11	10
12	(NC)
13	(NC)
14	(NC)



TI debug connection scheme

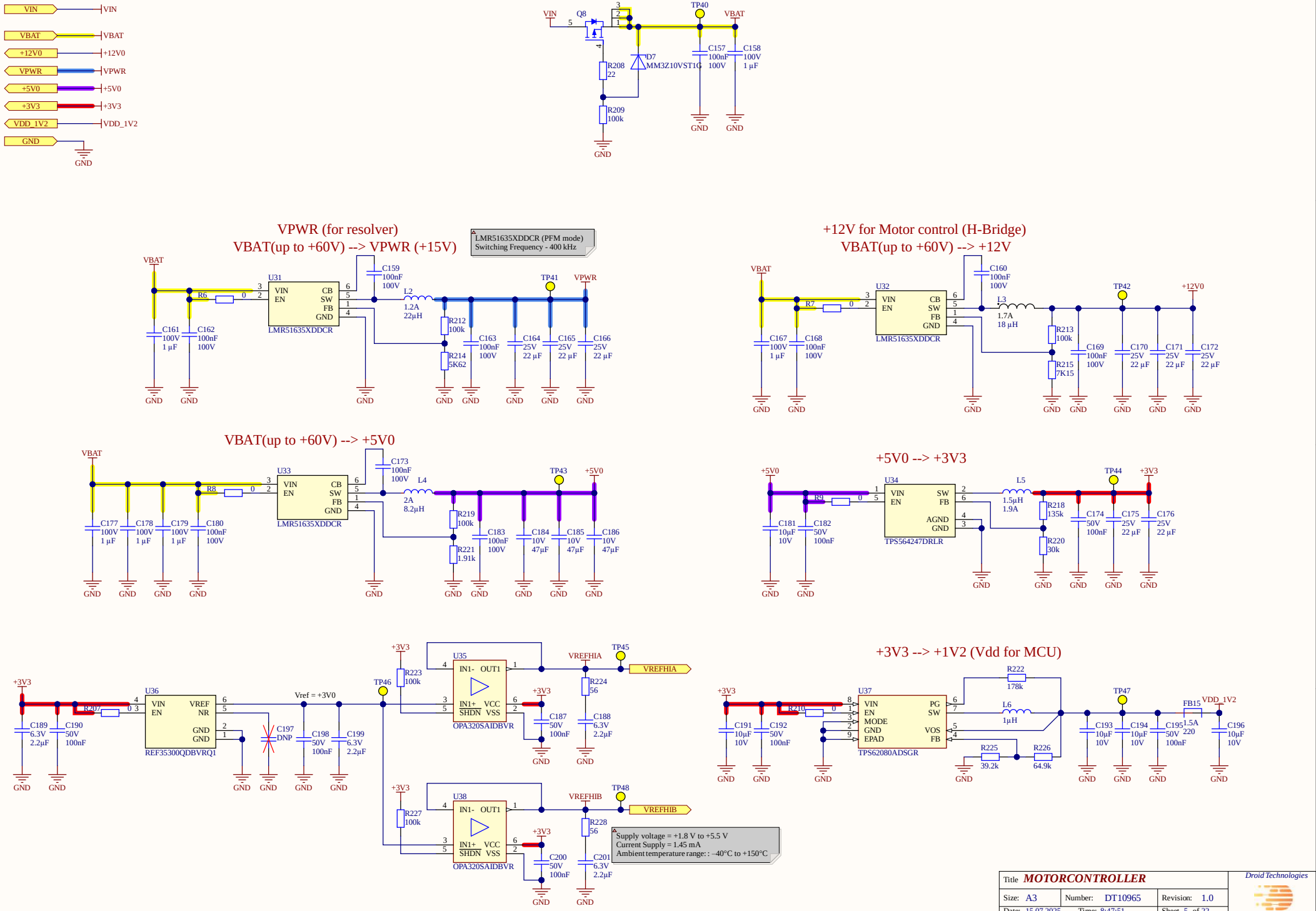
Figure 1. 14-Pin Header Signals and Header Dimensions

TMS	1	2	$\overline{\text{TRST}}$
TDI	3	4	GND
PD (V_{CC})	5	6	no pin (key)†
TDO	7	8	GND
TCK_RET	9	10	GND
TCK	11	12	GND
EMU0	13	14	EMU1

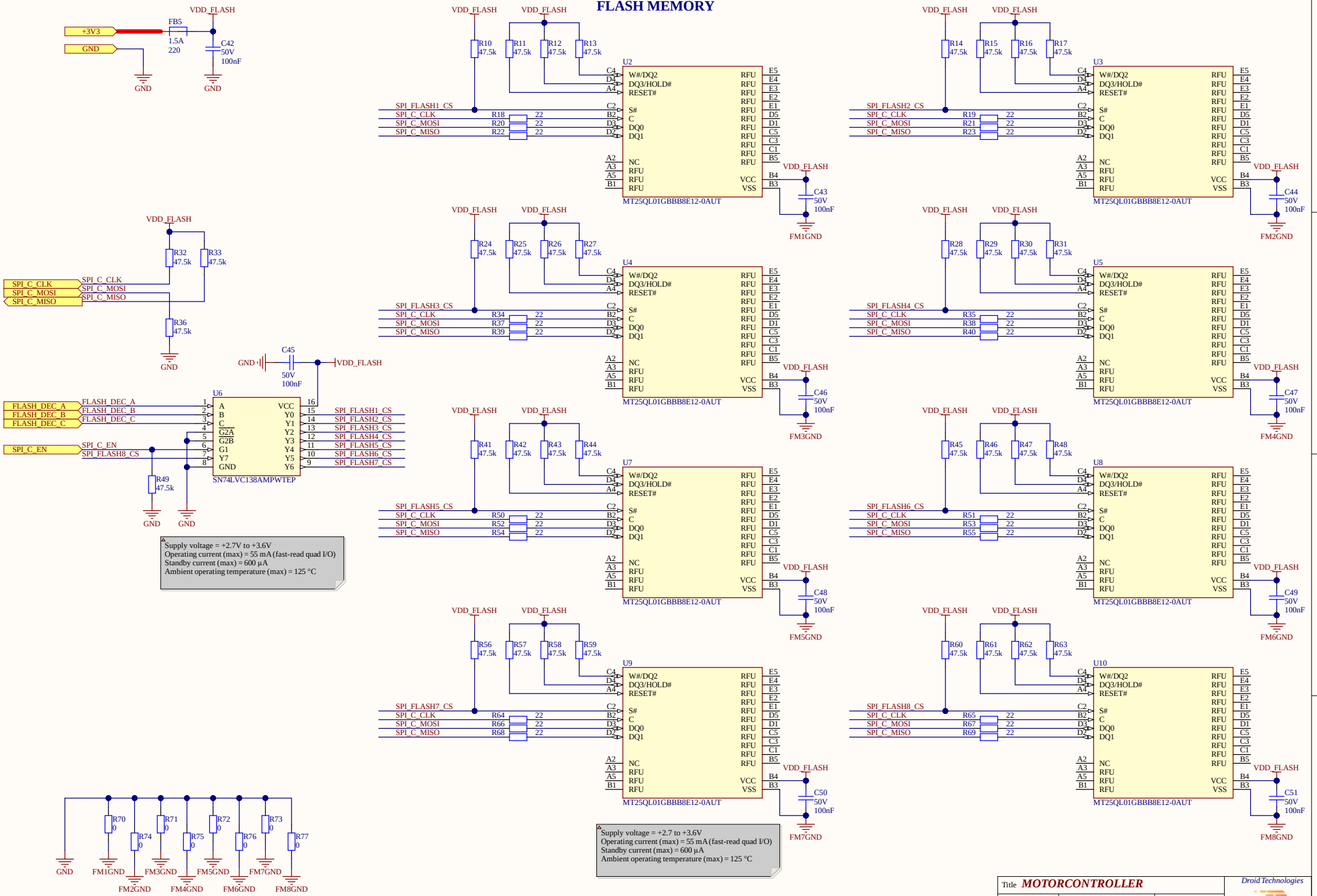
Header Dimensions:
Pin-to-pin spacing, 0.100 in. (X,Y)
Pin width, 0.025-in. square post
Pin length, 0.235-in. nominal

More details about debug connection from TI

Title	***MOTORCONTROLLER***		*Droid Technologies*
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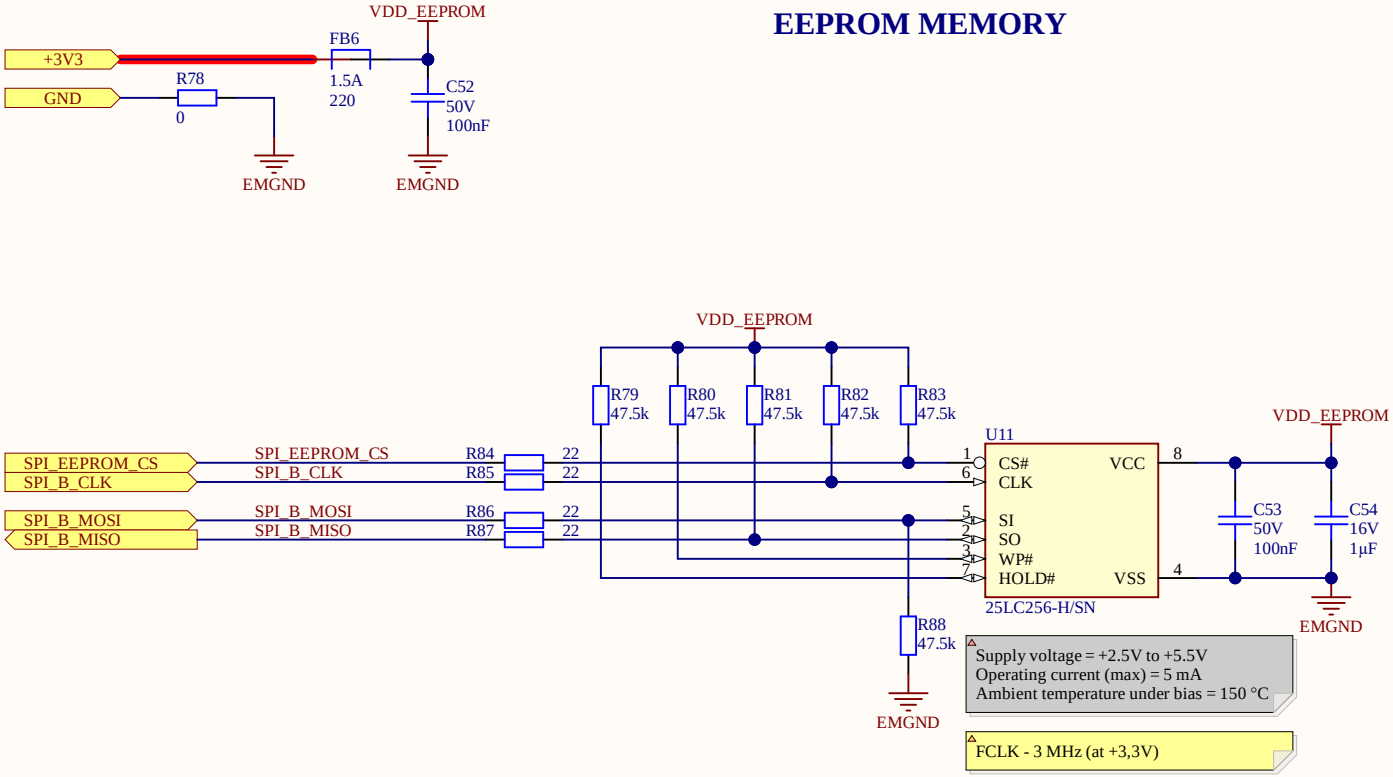

VIN
VBAT
+12V0
VPWR
+5V0
+3V3
VDD_1V2
GND
Q8
D7
MM3Z10VST1G
R208
22
R209
100k
TP40
C157
100nF
100V
C158
100V
1 µF
VPWR (for resolver)
VBAT(up to +60V) --> VPWR (+15V)
LMR51635XDDCR (PFM mode)
Switching Frequency - 400 kHz
U31
LMR51635XDDCR
L2
1.2A
22µH
TP41
R212
100k
R214
5K62
C159
C161
C162
C163
C164
C165
C166
25V
22 µF
+12V for Motor control (H-Bridge)
VBAT(up to +60V) --> +12V
U32
LMR51635XDDCR
L3
1.7A
18 µH
TP42
R213
100k
R215
7K15
C160
C167
C168
C169
C170
C171
C172
+12V0
VBAT(up to +60V) --> +5V0
U33
LMR51635XDDCR
L4
2A
8.2µH
TP43
+5V0
R219
100k
R221
1.91k
C173
C177
C178
C179
C180
C183
C184
C185
C186
10V
47µF
+5V0 --> +3V3
U34
TPS564247DRLR
L5
1.5µH
1.9A
TP44
+3V3
R218
135k
R220
30k
C174
C175
C176
C181
10µF
10V
C182
50V
100nF
+3V3 --> +1V2 (Vdd for MCU)
U37
TPS62080ADSGR
R210
R222
178k
L6
1µH
TP47
FB15
1.5A
220
VDD_1V2
R225
39.2k
R226
64.9k
C191
C192
C193
C194
C195
C196
U36
REF35300QDBVRQ1
Vref = +3V0
C197
DNP
C198
C199
C189
6.3V
2.2µF
C190
TP46
U35
OPA320SAIDBVR
R223
100k
TP45
VREFHIA
R224
56
C187
C188
U38
R227
100k
TP48
VREFHIB
R228
56
C200
C201
Supply voltage = +1.8 V to +5.5 V
Current Supply = 1.45 mA
Ambient temperature range: : –40°C to +150°C
Title MOTORCONTROLLER
Size: A3
Number: DT10965
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FLASH MEMORY



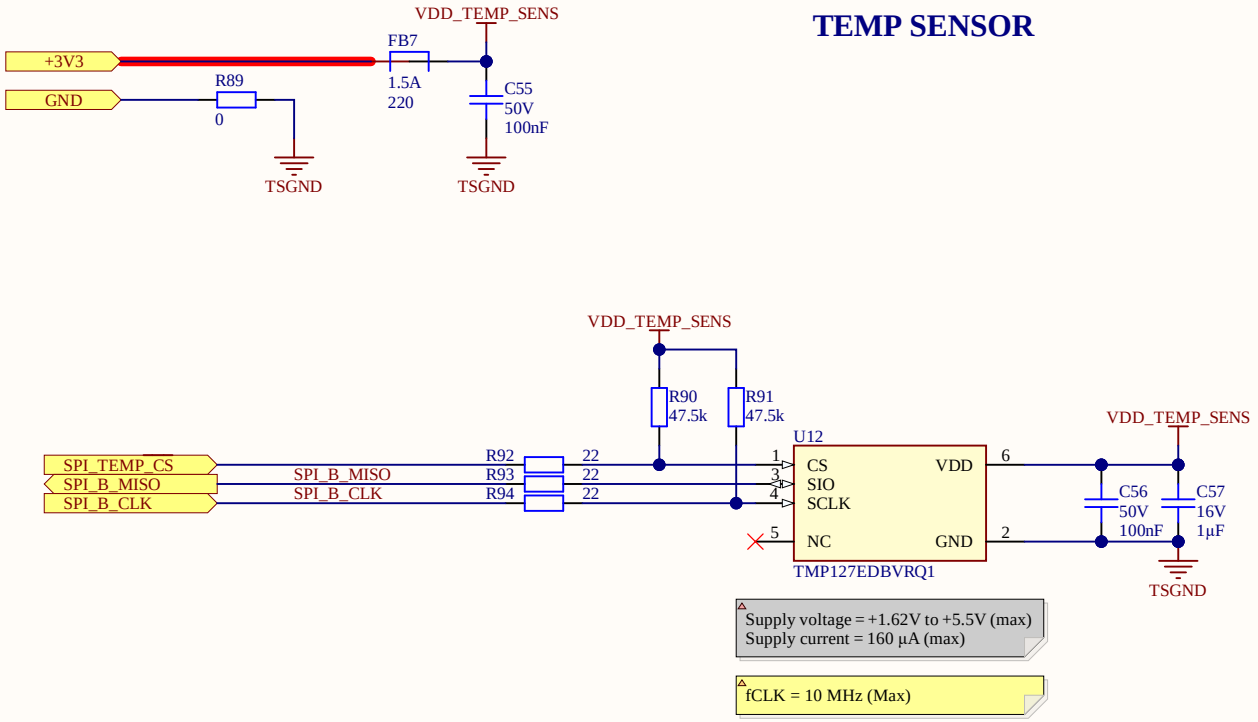
Title	MOTORCONTROLLER		Droid Technologies
Size: A3	Number: DT10965	Revision: 1.0	
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EEPROM MEMORY



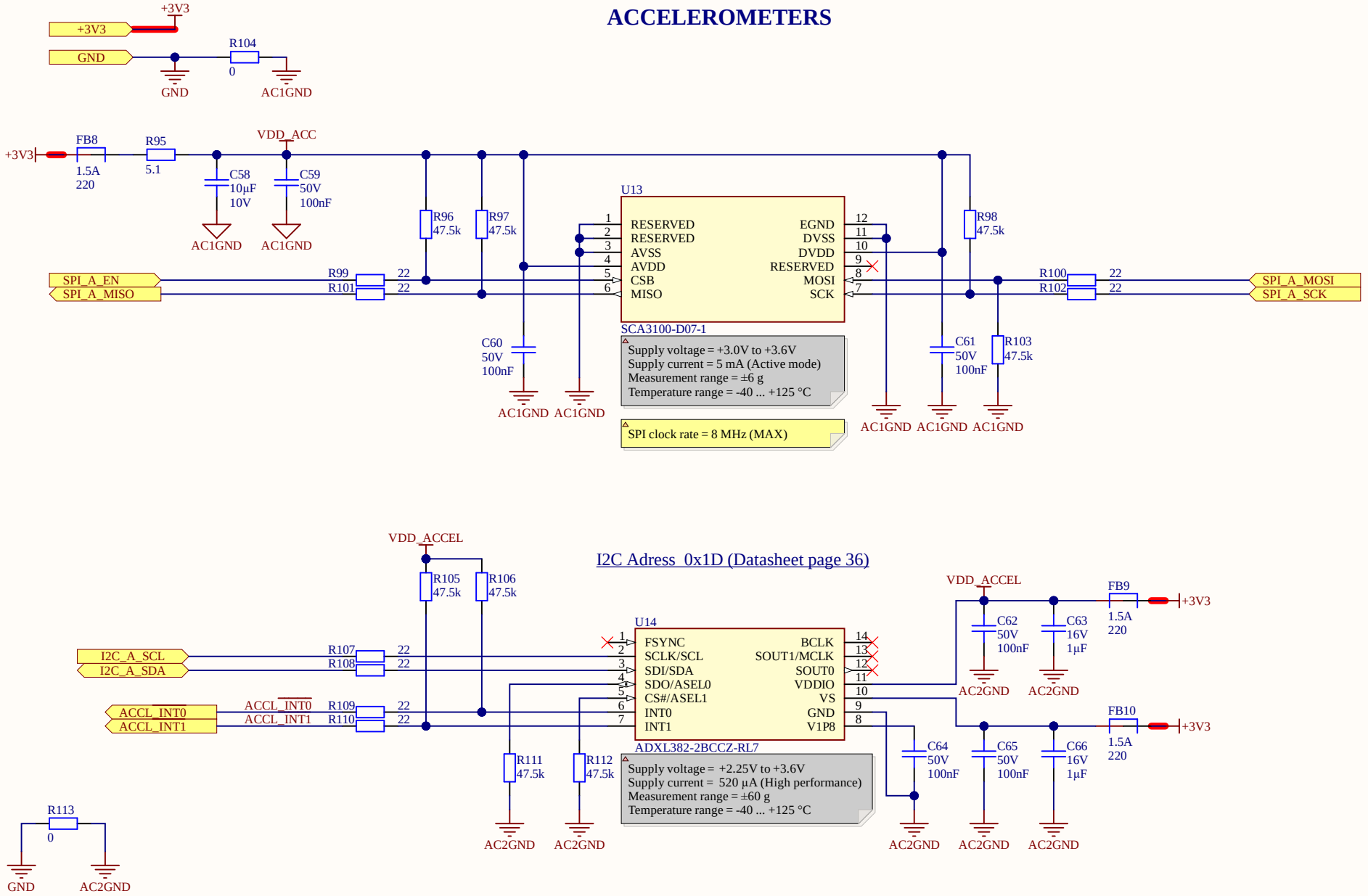
Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
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TEMP SENSOR

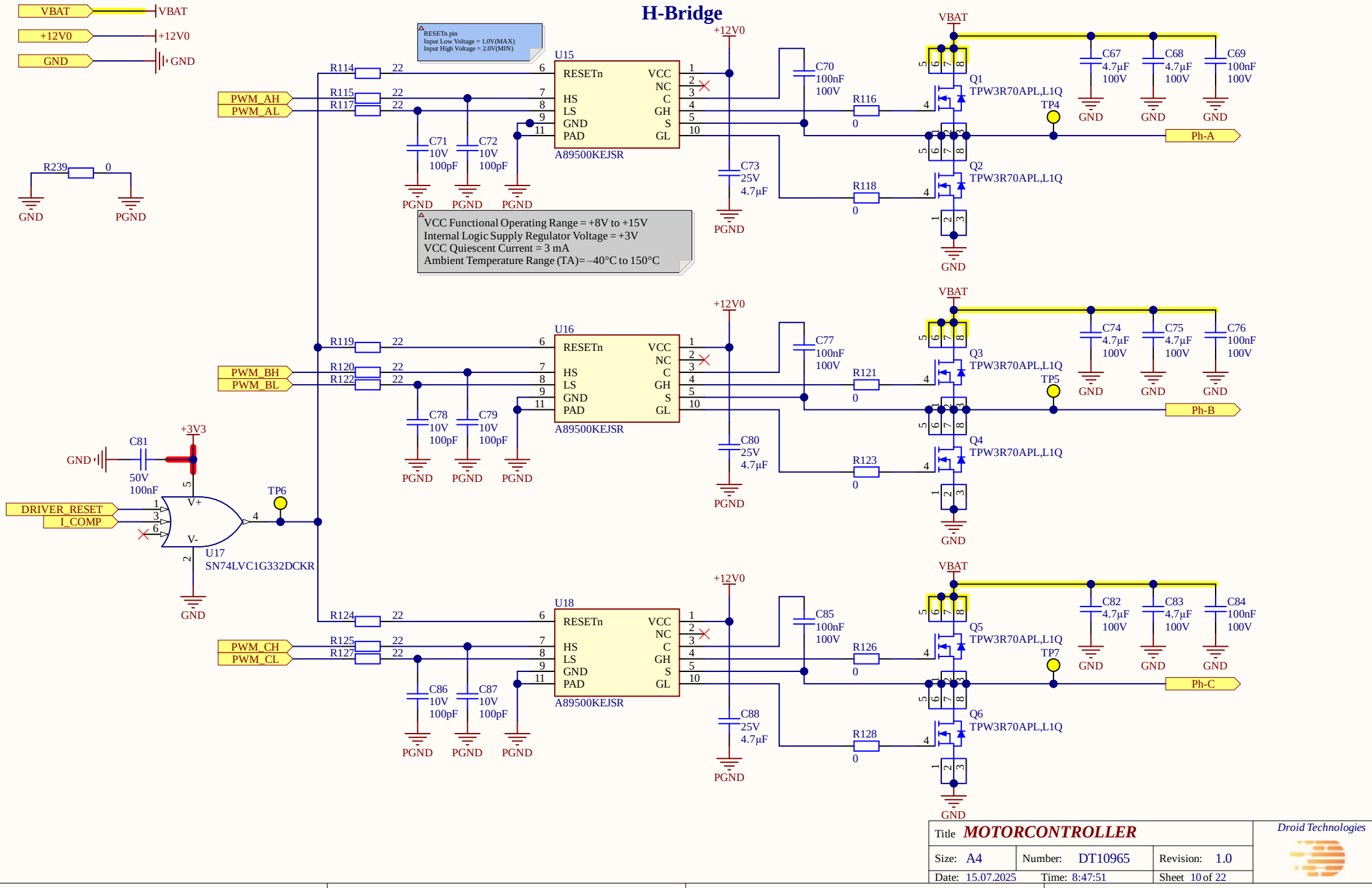


Title	***MOTORCONTROLLER***		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
Date: 15.07.2025	Time: 8:47:51	Sheet 8 of 22	

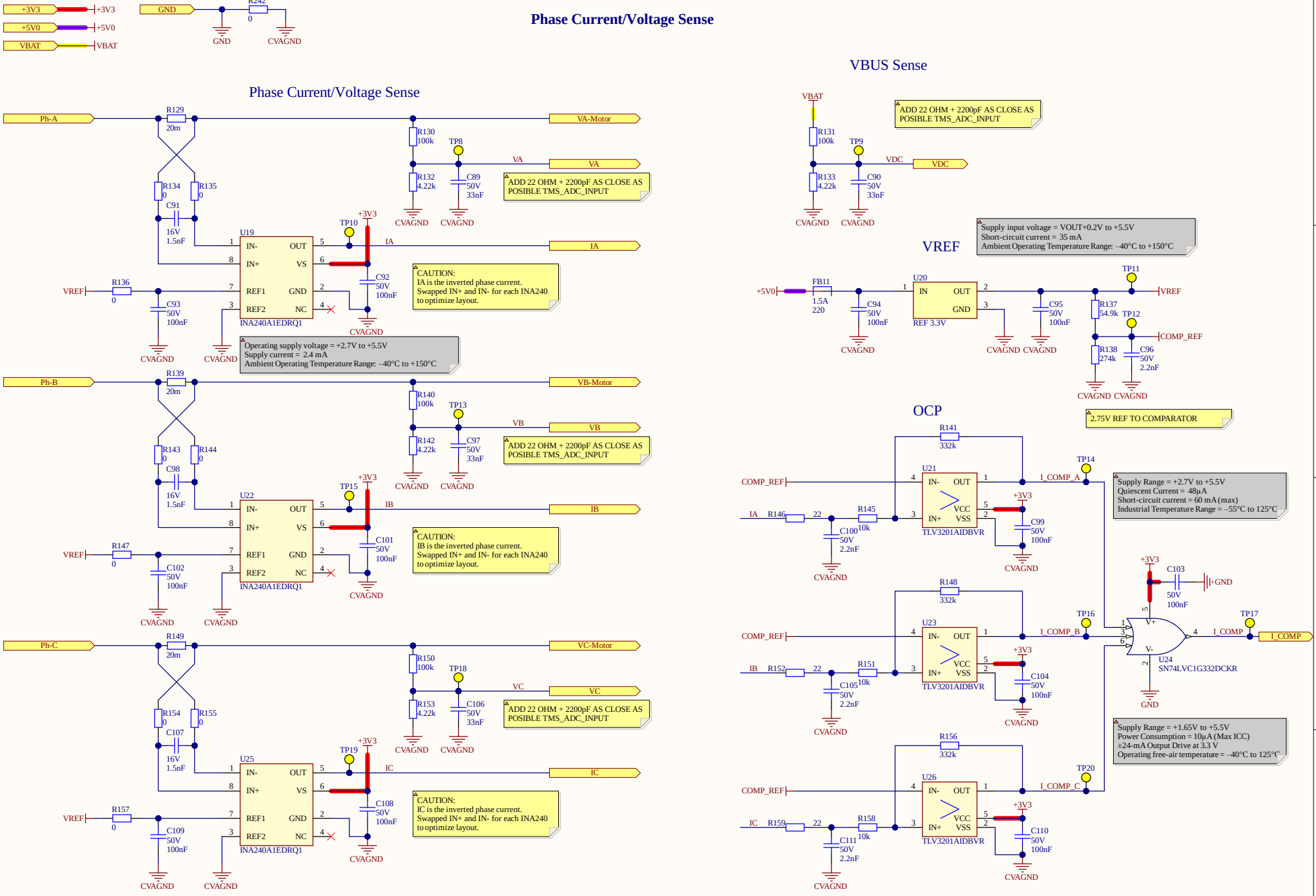
ACCELEROMETERS



Title	**MOTORCONTROLLER**		Droid Technologies
Size: A4	Number: DT10965	Revision: 1.0	
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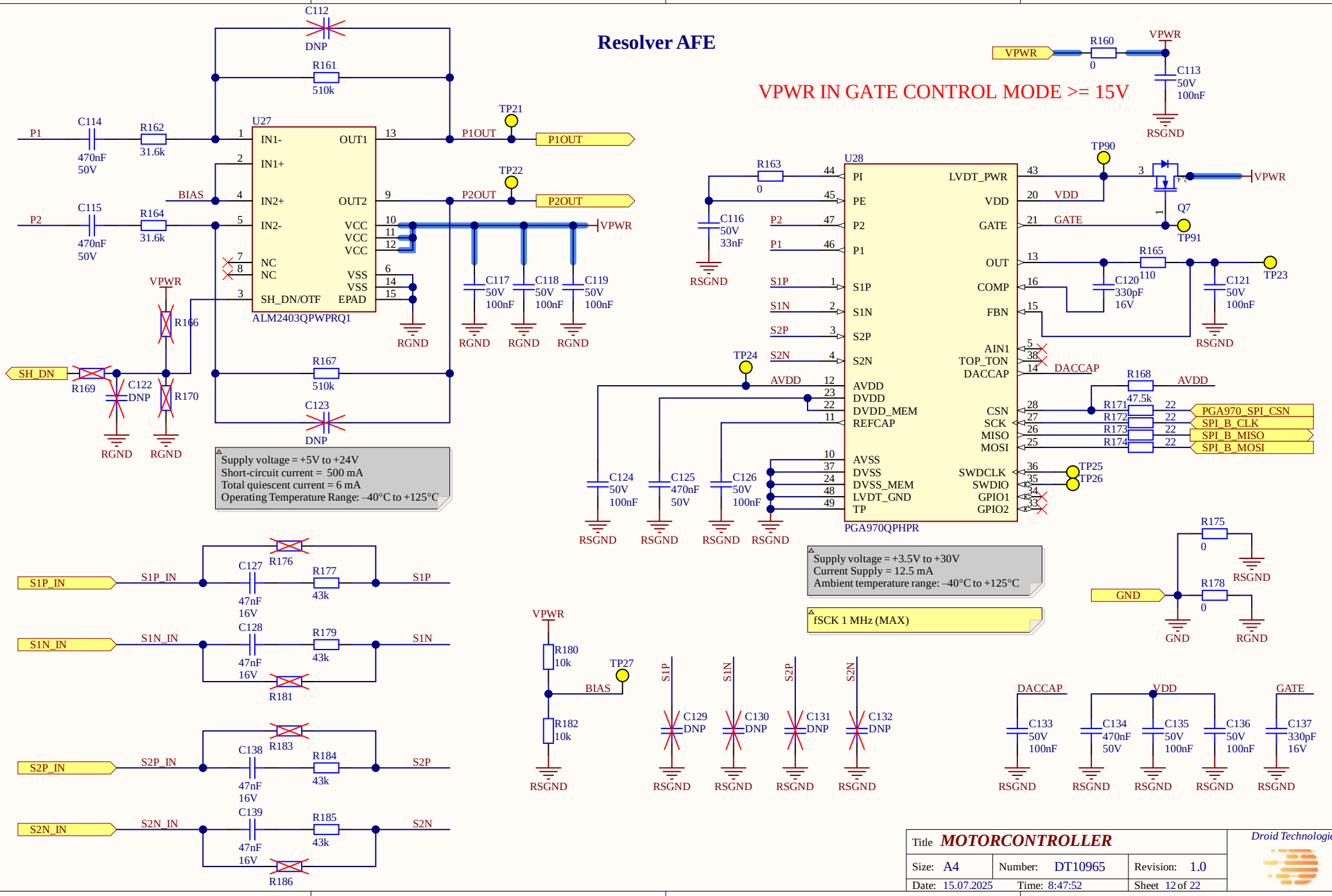

H-Bridge
VBAT
+12V0
GND
RESETn pin
Input Low Voltage = 1.0V(MAX)
Input High Voltage = 2.0V(MIN)
VCC Functional Operating Range = +8V to +15V
Internal Logic Supply Regulator Voltage = +3V
VCC Quiescent Current = 3 mA
Ambient Temperature Range (TA)= –40°C to 150°C
U15
U16
U18
A89500KEJSR
RESETn
HS
LS
GND
PAD
VCC
NC
C
GH
S
GL
PWM_AH
PWM_AL
PWM_BH
PWM_BL
PWM_CH
PWM_CL
R114
R115
R117
R119
R120
R122
R124
R125
R127
22
R239
0
C71
C72
C78
C79
C86
C87
10V
100pF
C70
C73
C77
C80
C85
C88
100nF
25V
4.7µF
100V
R116
R118
R121
R123
R126
R128
Q1
Q2
Q3
Q4
Q5
Q6
TPW3R70APL,L1Q
C67
C68
C69
C74
C75
C76
C82
C83
C84
TP4
TP5
TP7
Ph-A
Ph-B
Ph-C
PGND
+3V3
C81
50V
100nF
TP6
DRIVER_RESET
I_COMP
U17
SN74LVC1G332DCKR
V+
V-
Title MOTORCONTROLLER
Size: A4
Number: DT10965
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Phase Current/Voltage Sense



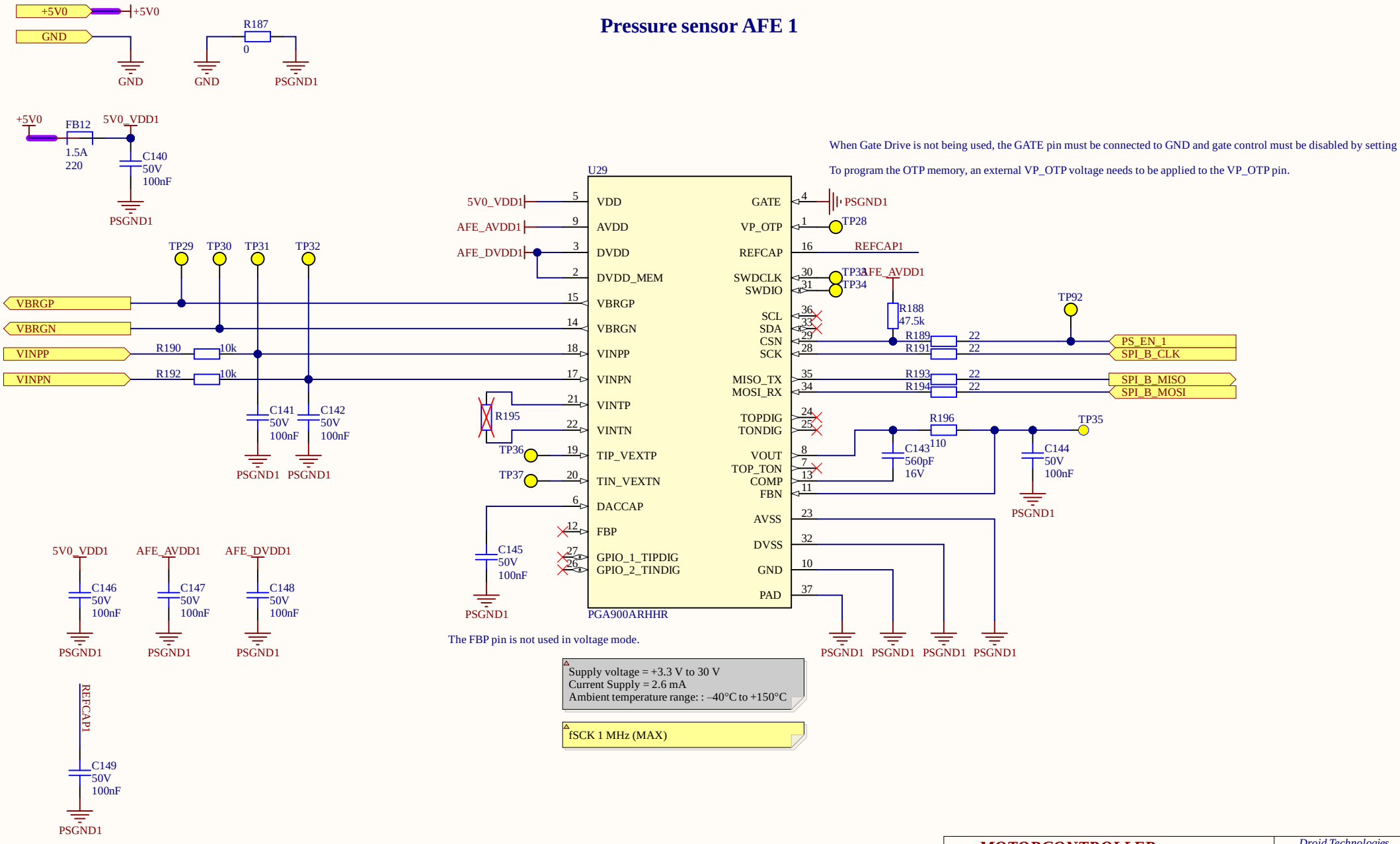
Title	MOTORCONTROLLER		Droid Technologies
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Resolver AFE



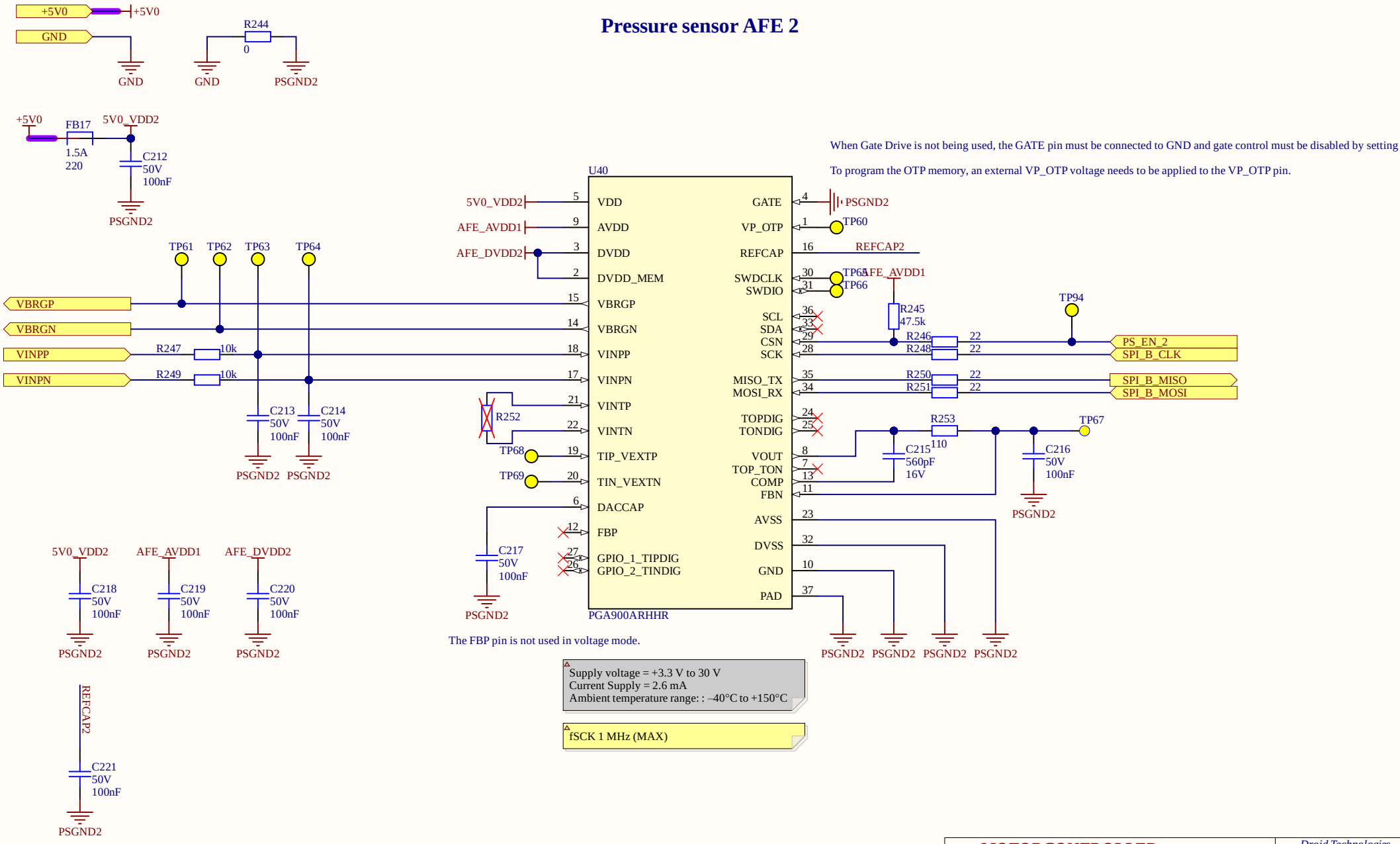
Title	**MOTORCONTROLLER**		*Droid Technologies*
Size: A4	Number: DT10965	Revision: 1.0	
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Pressure sensor AFE 1



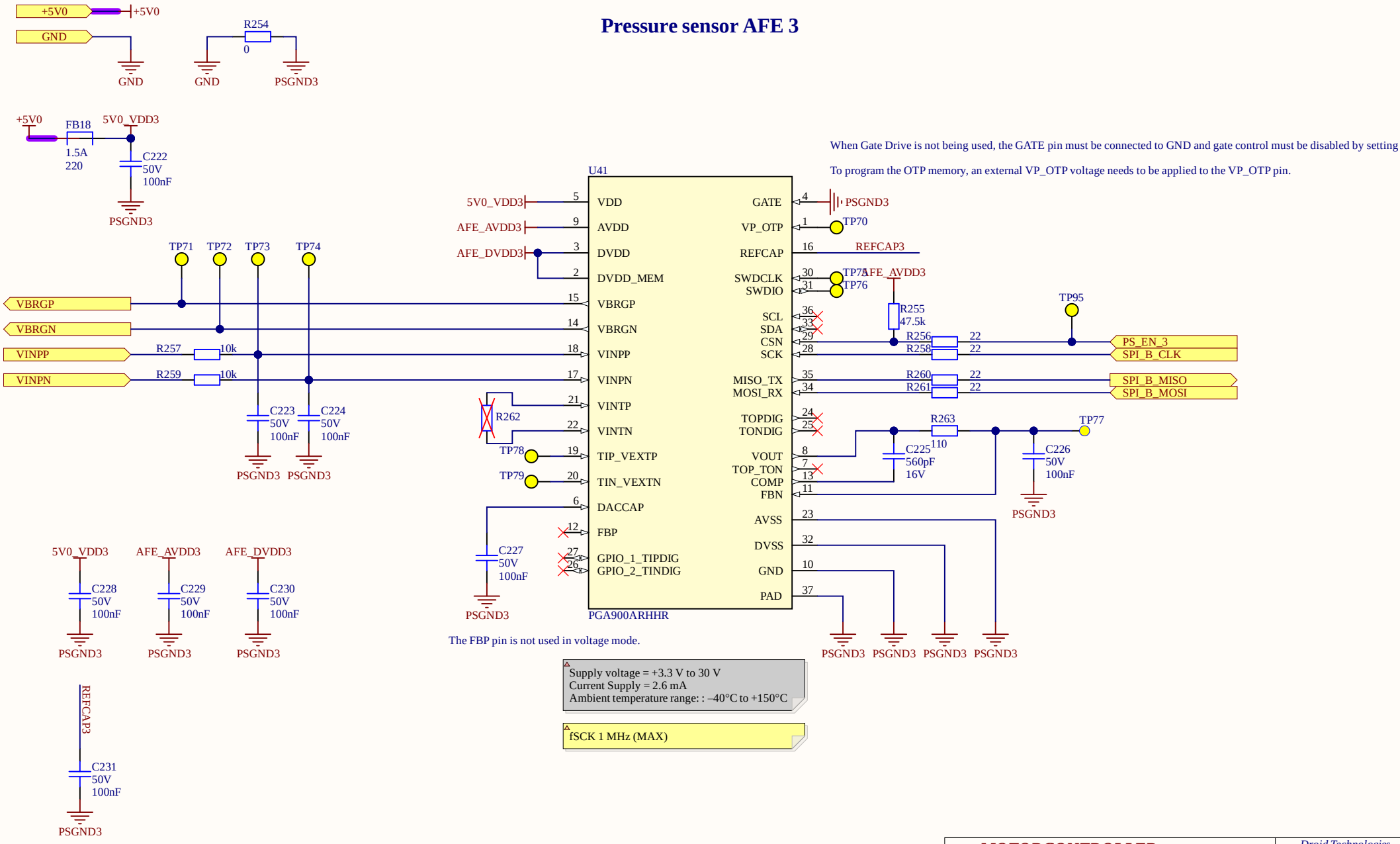
Title	**MOTORCONTROLLER**		*Droid Technologies*
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Pressure sensor AFE 2



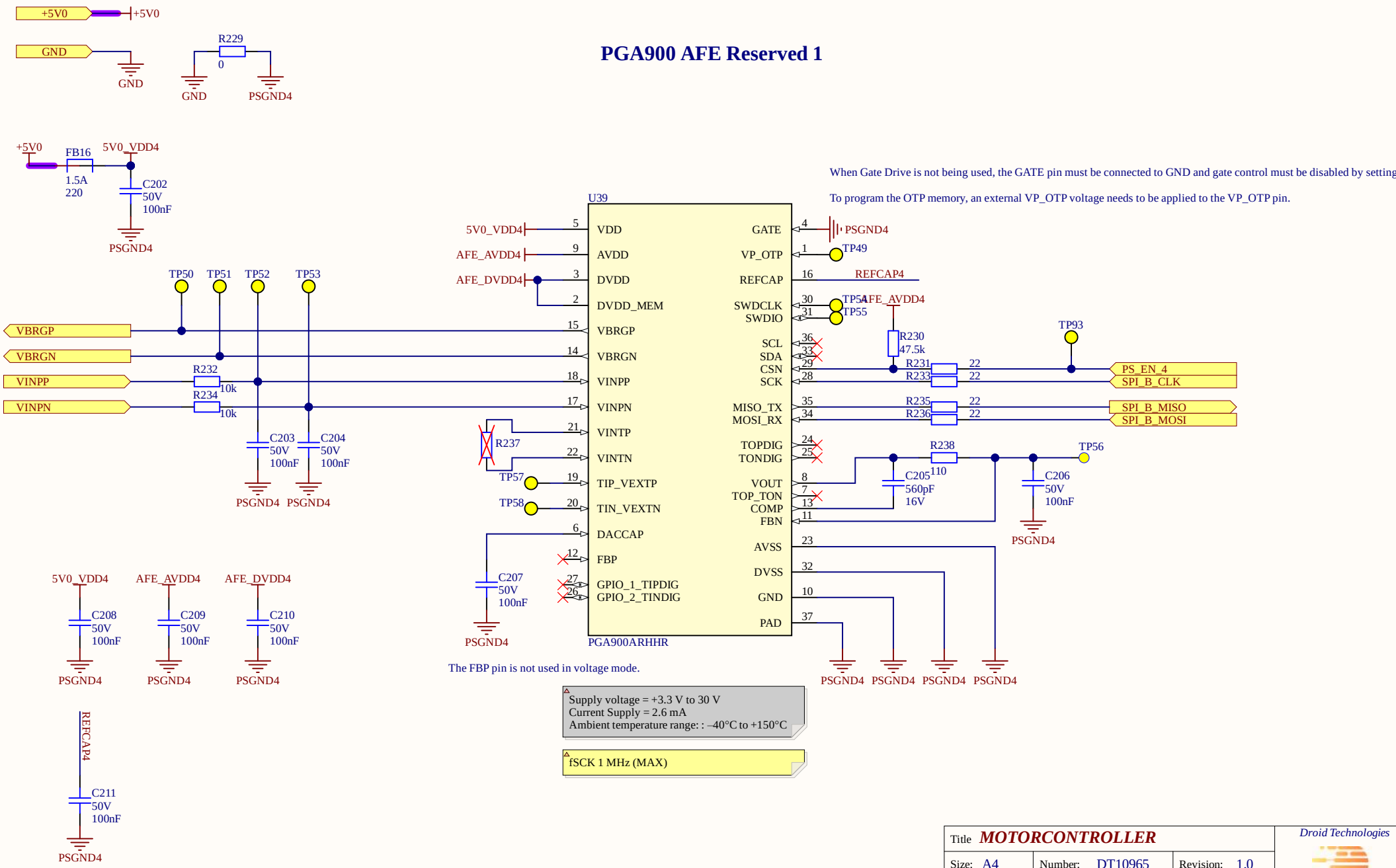
Title	**MOTORCONTROLLER**		*Droid Technologies*
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Pressure sensor AFE 3



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PGA900 AFE Reserved 1



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PGA900 AFE Reserved 2

+5V0 +5V0

GND GND

R264 0 GND PSGND5

+5V0 FB19 1.5A 220 5V0_VDD5 C232 50V 100nF PSGND5

TP81 TP82 TP83 TP84

VBRGP VBRGN VINPP VINPN

R267 10k R269 10k

C233 50V 100nF C234 50V 100nF PSGND5 PSGND5

U42

5V0_VDD5	5 VDD	GATE 4	PSGND5
AFE_AVDD5	9 AVDD	VP_OTP 1	TP80
AFE_DVDD5	3 DVDD	REFCAP 16	REFCAP5
	2 DVDD_MEM	SWDCLK 30 / SWDIO 31	TP85 TP86
	15 VBRGP	SCL 36 / SDA 33 / CSN 29 / SCK 28	
	14 VBRGN	MISO_TX 35 / MOSI_RX 34	
	18 VINPP	TOPDIG 24 / TONDIG 25	
	17 VINPN	VOUT 8 / TOP_TON 7 / COMP 13 / FBN 11	
	21 VINTP	AVSS 23	
	22 VINTN	DVSS 32	
	19 TIP_VEXTP	GND 10	
	20 TIN_VEXTN	PAD 37	
	6 DACCAP		
	12 FBP		
	27 GPIO_1_TIPDIG / 26 GPIO_2_TINDIG		

PGA900ARHHR

R272

TP88 TP89

C237 50V 100nF PSGND5

AFE_AVDD5 R265 47.5k

R266 22 R268 22 PS_EN_5 SPI_B_CLK

R270 22 R271 22 SPI_B_MISO SPI_B_MOSI

TP96

R273 110 C235 560pF 16V C236 50V 100nF TP87 PSGND5

PSGND5 PSGND5 PSGND5 PSGND5

5V0_VDD5 C238 50V 100nF PSGND5

AFE_AVDD5 C239 50V 100nF PSGND5

AFE_DVDD5 C240 50V 100nF PSGND5

REFCAP5 C241 50V 100nF PSGND5

When Gate Drive is not being used, the GATE pin must be connected to GND and gate control must be disabled by setting

To program the OTP memory, an external VP_OTP voltage needs to be applied to the VP_OTP pin.

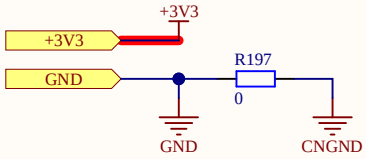
The FBP pin is not used in voltage mode.

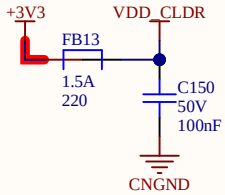
Supply voltage = +3.3 V to 30 V
Current Supply = 2.6 mA
Ambient temperature range: : –40°C to +150°C

fSCK 1 MHz (MAX)

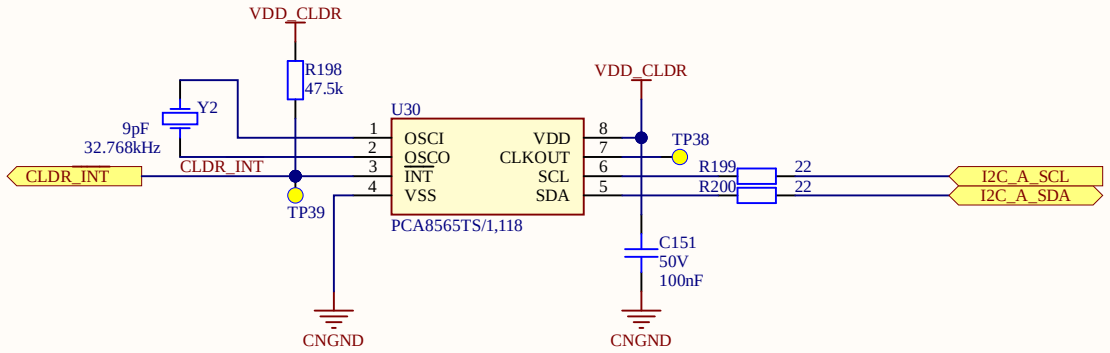
Title	MOTORCONTROLLER		Droid Technologies
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CALENDAR





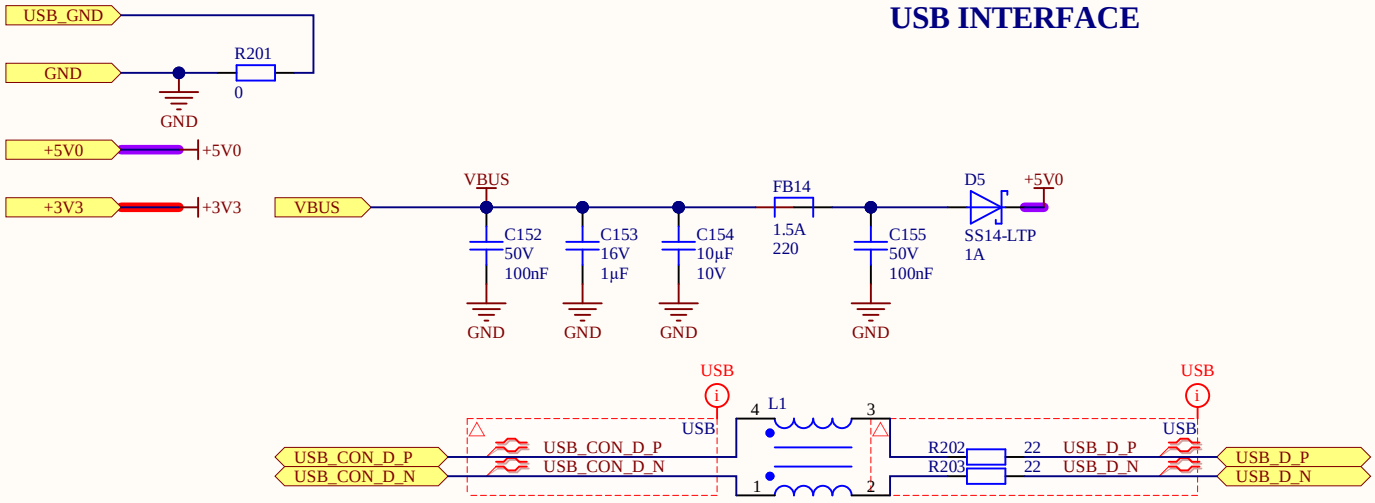
I2C Adresses 0x00 to 0x0F, 0xA2, 0xA3 (Datasheet page 8)

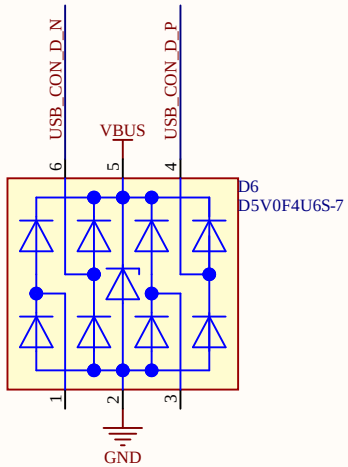


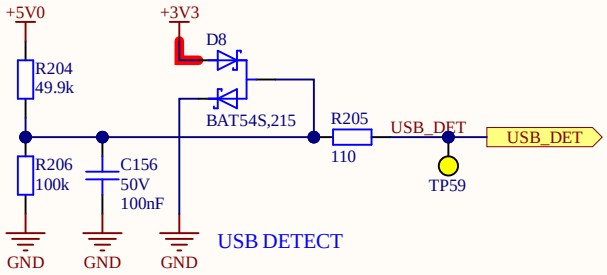
Supply voltage = : 0.9 V to 5.5 V
Current Supply = 820 µA
Ambient temperature range: : –40°C to +150°C

Title **MOTORCONTROLLER**			*Droid Technologies*
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USB INTERFACE







Any pin that has digital input and output functionality can potentially be used to monitor V_{Bus}, but the use of an interrupt-capable GPIO is recommended. A pin that does not have external interrupt capability can also be used, but the input state of the pin must be polled periodically by the application software to make sure appropriate action is taken whenever V_{Bus} is applied or removed. If an interrupt-capable GPIO is chosen, the GPIO can be configured to generate an interrupt on both the rising and falling edge. More information on external interrupts can be found in the *System Control and Interrupts* chapter. Example code that implements V_{Bus} monitoring using external interrupts and takes the appropriate actions is documented in the USB Software Guide and can be found in the associated USB software package.

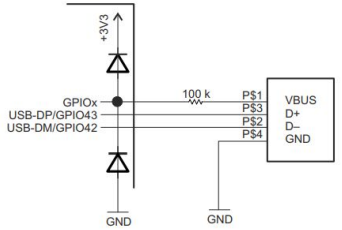
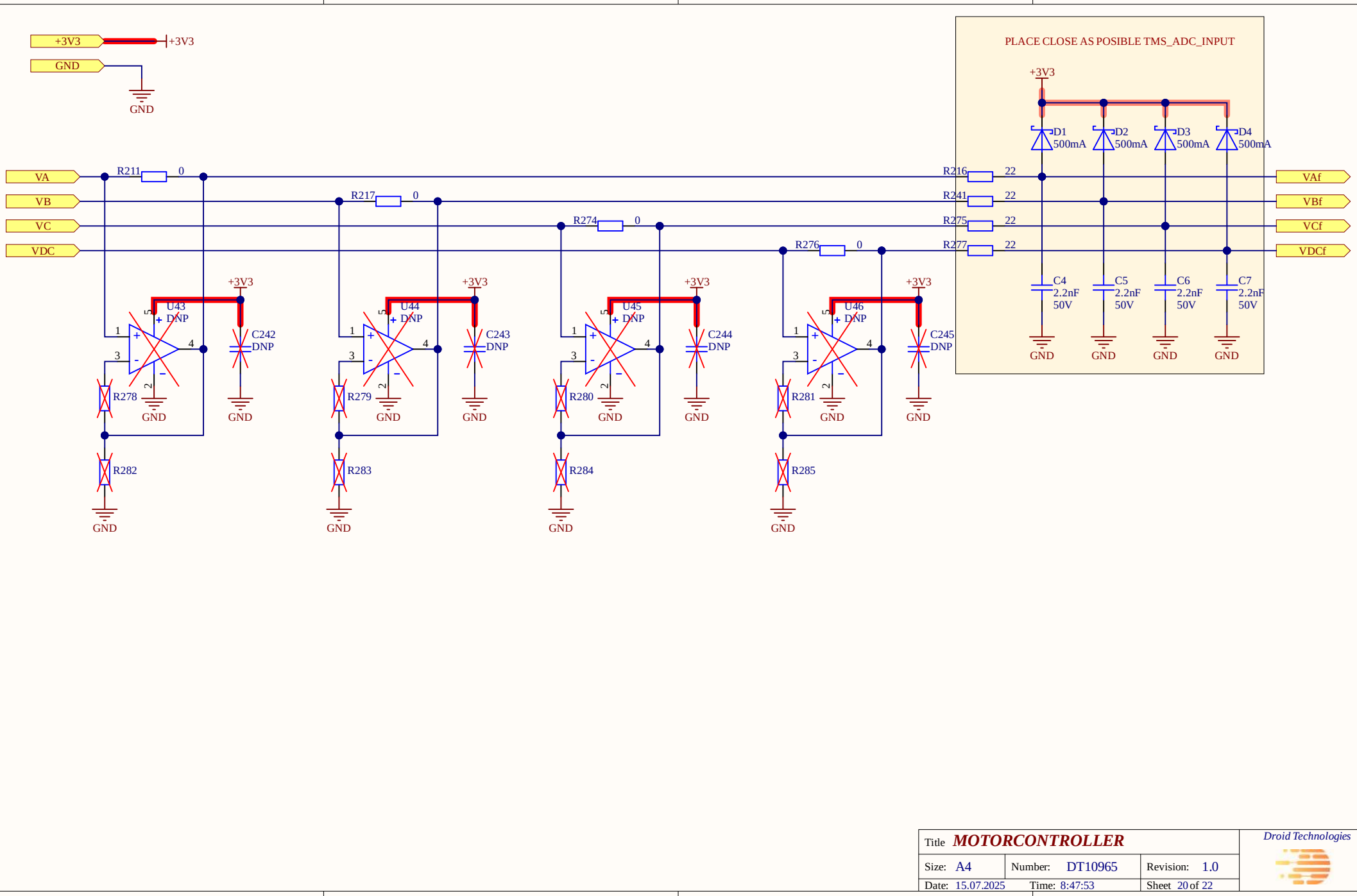


Figure 22-2. USB Scheme

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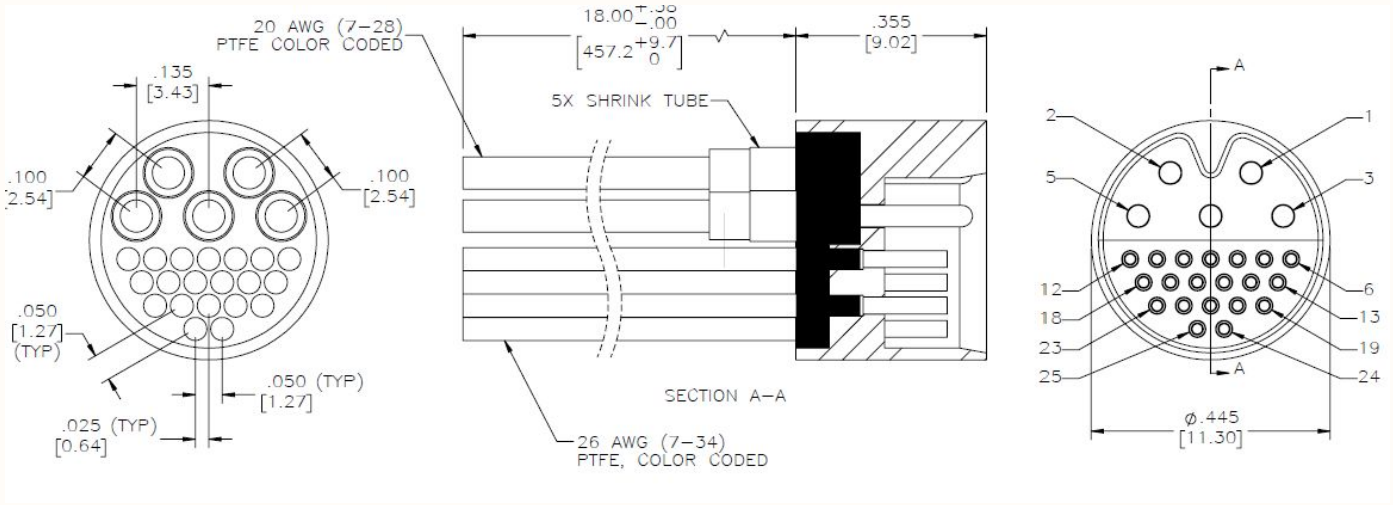


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J2 Wiring connection scheme

AWG 20 wires				
Contact #	Color	Signal	External equipment	PCB connection
1	Black	GND	Battery	X32
2	Brown	n.a.	n.a.	n.a.
3	Red	Vin	Battery	X31
4	Orange	n.a.	n.a.	n.a.
5	Yellow	n.a.	n.a.	n.a.

AWG 26 wires				
Contact #	Color	Signal	External equipment	PCB connection
6		VBRGP1	Pressure sensor1	X11
7		VBRGN1	Pressure sensor1	X12
8	Red	VINPP1	Pressure sensor1	X13
9	Orange	VINPN1	Pressure sensor1	X14
10	Yellow	VBRGP2	Pressure sensor2	X15
11	Green	VBRGN2	Pressure sensor2	X16
12	Blue	VINPP2	Pressure sensor2	X17
13		VINPN2	Pressure sensor2	X18
14	Gray	VBRGP3	Pressure sensor3	X19
15	White	VBRGN3	Pressure sensor3	X20
16		VINPP3	Pressure sensor3	X21
17		VINPN3	Pressure sensor3	X22
18	Red	VBRGP4	Pressure sensor4	X23
19	Orange	VBRGN4	Pressure sensor4	X24
20	Yellow	VINPP4	Pressure sensor4	X25
21	Green	VINPN4	Pressure sensor4	X26
22	Blue	VBRGP5	Pressure sensor5	X27
23		VBRGN5	Pressure sensor5	X28
24	Gray	VINPP5	Pressure sensor5	X29
25	White	VINPN5	Pressure sensor5	X30

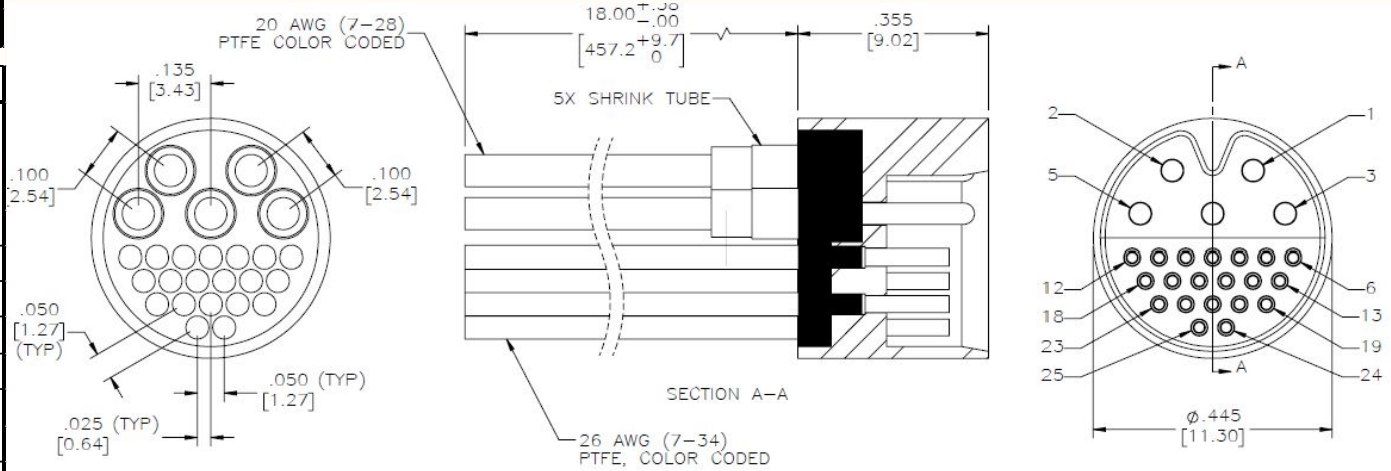


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J3 Wiring connection scheme

AWG 20 wires				
Contact #	Color	Signal	External equipment	PCB connection
1	Black	n.a.	n.a.	n.a
2	Brown	Winding 1	Motor	X33.
3	Red	n.a.	n.a.	n.a.
4	Orange	Winding 2	Motor	X34
5	Yellow	Winding 3	Motor	X35

AWG 26 wires				
Contact #	Color	Signal	External equipment	PCB connection
6	Green	P1OUT	Resolver	X1
7	Blue	P2OUT	Resolver	X2
8	Red	S1P_IN	Resolver	X3
9	Orange	S1N_IN	Resolver	X4
10	Yellow	S2P_IN	Resolver	X5
11	Green	S2N_IN	Resolver	X6
12	Blue	n.a.	n.a.	n.a.
13	Violet	n.a.	n.a.	n.a.
14	Gray	n.a.	n.a.	n.a.
15	White	n.a.	n.a.	n.a.
16	Black	USB_GND	USB	X10
17	Brown	n.a.	n.a.	n.a.
18	Red	VBUS	USB	X9
19	Orange	USB_CON_D_P	USB	X7
20	Yellow	USB_CON_D_N	USB	X8
21	Green	n.a.	n.a.	n.a.
22	Blue	n.a.	n.a.	n.a.
23	Violet	n.a.	n.a.	n.a.
24	Gray	n.a.	n.a.	n.a.
25	White	n.a.	n.a.	n.a.



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